CHEZHIAN ARTS AND SCIENCE COLLEGE FOR WOMEN

THENPALLIPATTU, KALASAPAKKAM TALUK THIRUVANNAMALAI

BACHELOR OF COMPUTER **SCIENCE**

NAAN MUDHALVAN FINALL PROJECT RECORD

PROJECT TITLE:

STUDENT ENROLLMENT AND COURSE MANAGEMENT SYSTEM

SUBMITTED BY:

TEAM LEADER : VISHNUPRIYA R 24723U18048

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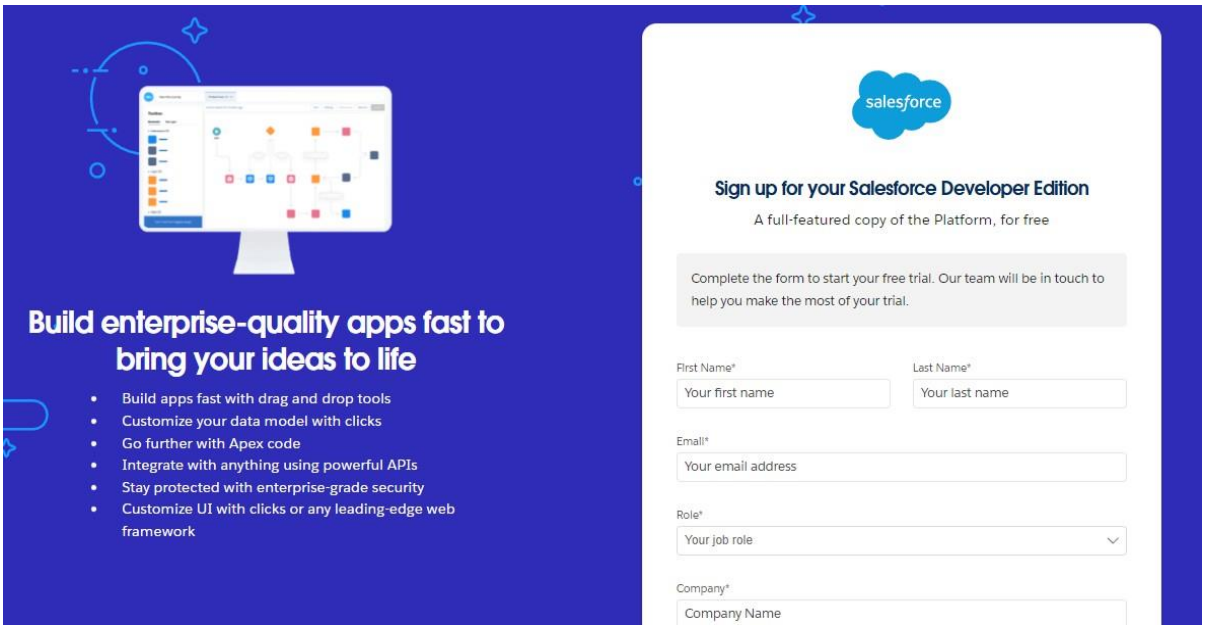
1.1 Project Overview :

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations—from admission to course completion—are handled systematically and with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

Activity 1: Creating Developer Account:

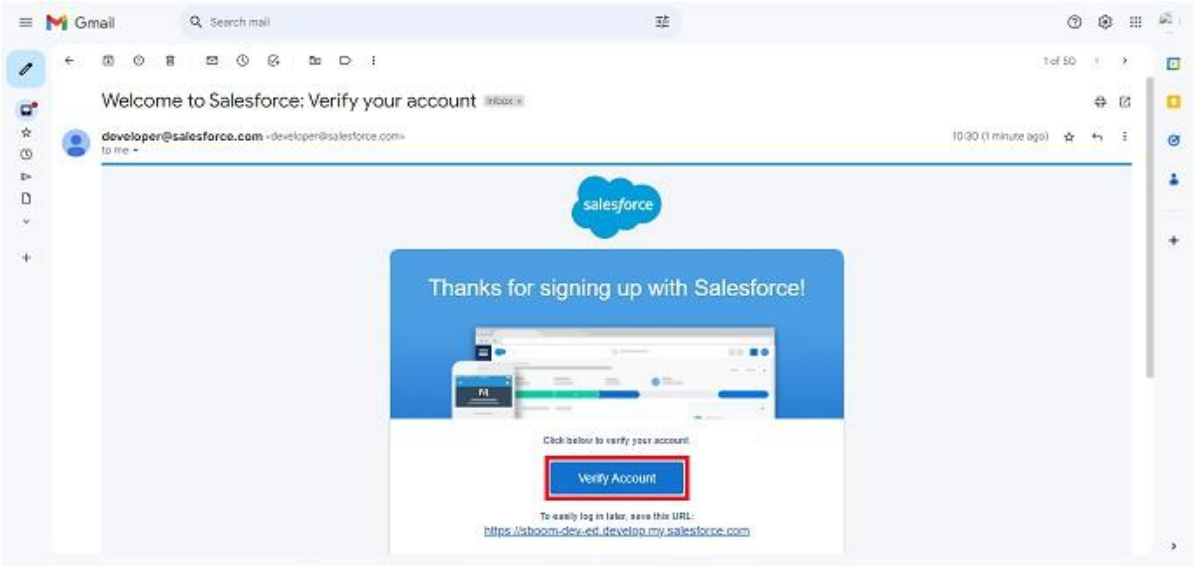
Creating a developer org in salesforce.
Go to https://developer.salesforce.com/signup
On the sign up form, enter the following details :



First name & Last name
Email
Role : Develop
Company : College Name
County : India
Postal Code : pin code
Username : should be a combination of your name and company
This need not be an actual email id, you can give anything in the format :
username@organization.com
Click on sign me up after filling these.

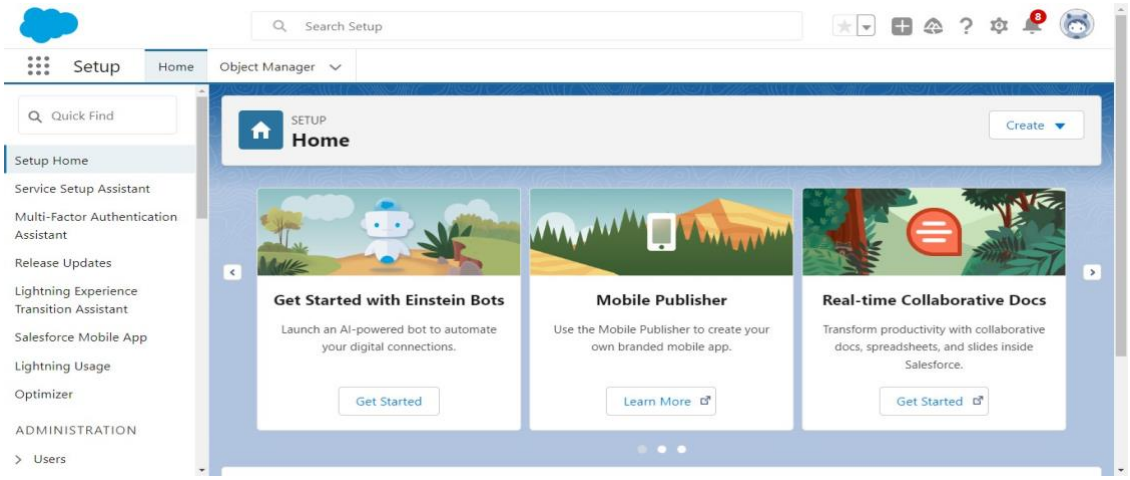
Activity 2: Account Activation:

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account
3. Give a password and answer a security question and click on change.

4. Then you will redirect to your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the Student object:

Go to the Setup page.

Click on Object

Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

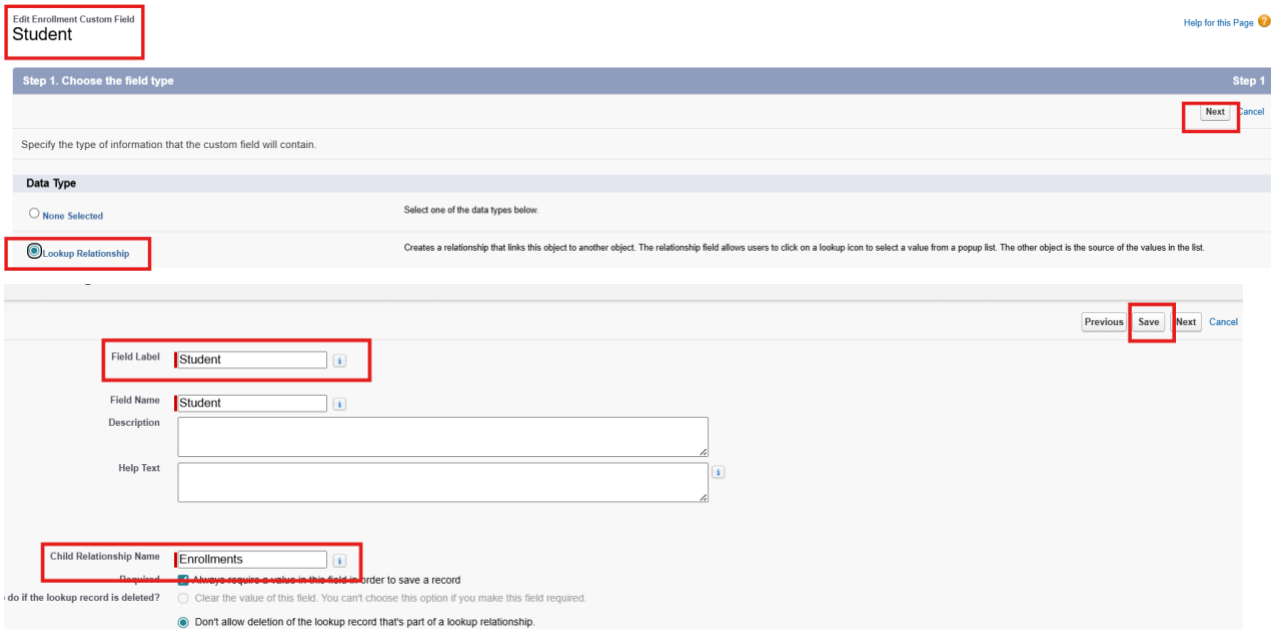
Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Create Relationships:

1. **Activity 1: Create Lookup from Enrollment to Student:**
   Go to Setup → click Object Manager.

   1. Type Enrollment in Quick Find → click Enrollment__c object.
   2. Click Fields & Relationships → click New.
   3. Select Lookup Relationship → click Next.
   4. For Related To, select Student__c → click Next.

5. Enter Field Label: Student (API Name: Student__c).
6. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).
7. Click Save.

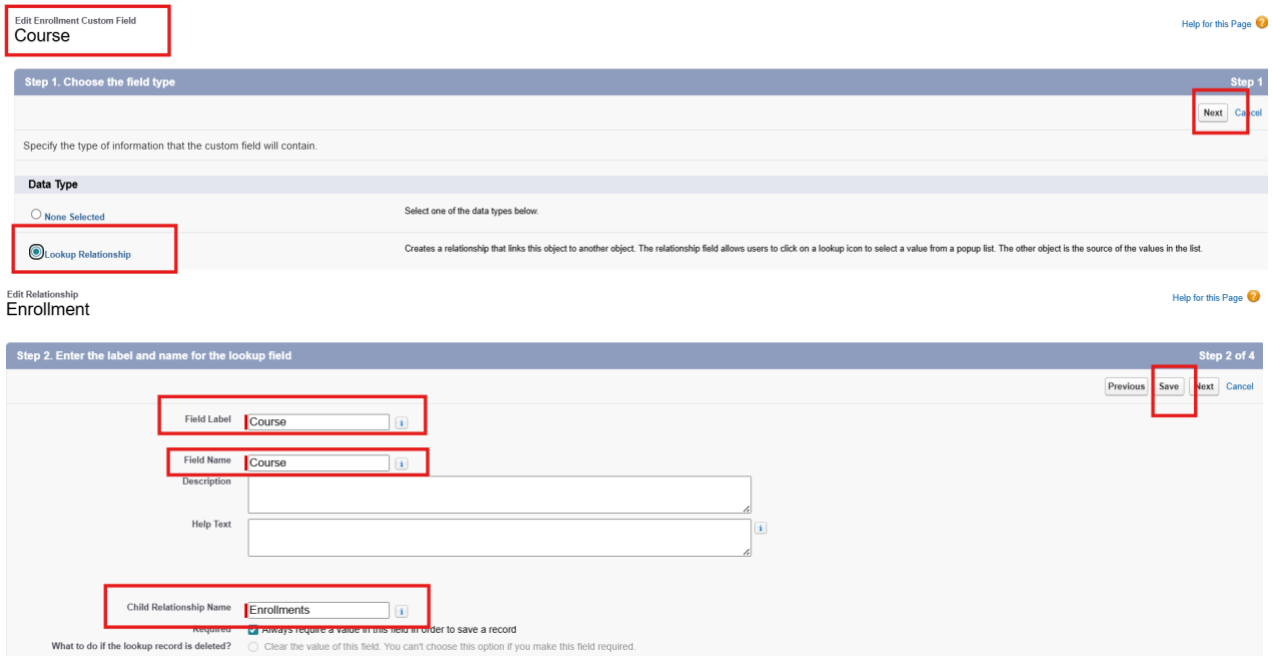


Activity 2: Create Lookup from Enrollment to Course

Purpose:
To link each Enrollment to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course__c → click Next.
5. Field Label: Course (API Name: Course__c).
6. Click Next (set visibility and layout) → Save.



Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:
To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor__c → Next.
5. Field Label: Instructor (API Name: Instructor__c).
6. Complete Field-Level Security and Page Layout options → Save

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons,Links,andActions

Compact Layouts

FieldSel5

Object limits

Specify the type of information that the custom field willcontain

Data Type

ONoneSelected — Selectorn!oltthedatatypesbelow_

OAutoNumber — Asystem-gooefated sequence mrmbef that uses a diSJlay format you define. The number is aLrtomatically incremented for each new record

0Formula — A read-ontyfield that derives its vakle from a formula expression you define. Theformula field is updated when any of the sourcefields char,ge

Roll-Up Summuy ...f... — A read-only field that cisplays the sum, minimim,()(mauflum vaUe of a field in a related list()(the record count of al rncords fisted in a r□ated list

@LookupRelationship — Creates a rwationshJpthat llks thisoqecl to another object The retatiooshipfietd allows users lo dick on a Wup icontoselectavalue from a popup isl. The other object is the soorceofthevaluesinlhelist

Enrollment

New Relationship

Help for this Page e

Step 2 Choose the related obJect — Step 2

Previous Next Cancel

Select the other object to which this oDject is related

Related To Ins_tru_c_to_r V

Previous Next Cancel

Step 3 Enter the label and name for the lookup field — Step J of 6

Previous Next Cancel

Field Label Instrudor

Field Name Instructor

Description

Help Text

Child Relationship Name Enrollments

Required r3 AW<! s require a value in this fteld in order to save a record

What to do if e o up reco 1s deleted? Clear the value of this field. You can't choose this option if you make this field required

@ Don't allow deletion of the lookup record that's part or a lookup relationship

Auto add to custom report ty□ r3 Add this rield to existing custom report types that contain lhis enlily i

Enrollment

New Relationship

HelpforlhisPilge

Step 6 Add custom related hsts — Step 6 of 6

iPrevious iSe &Newlb tanool

Field label Instructor

Data Ty□ Lookup

Field Name Instructor

Description

Specify the title that llle related list Will have in all of the layouts associated With the parent.

Related List Label E_n_ro_llm_e_n_ts

Select the page layouts that should include this field. The field will be added as the last field in the first 2--column section of these page layouts_ The field will not appear on any pages if you do not select a layoul

To change the location of this field on the page, you Will need to customize the page layout To change the values that appear, you Will need to customiZe the Record Types

Add Related Lisl Page Layout Nnme

£!3 Instructor LayoUl

rJ Append related list to users' existing personal customizalions

Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

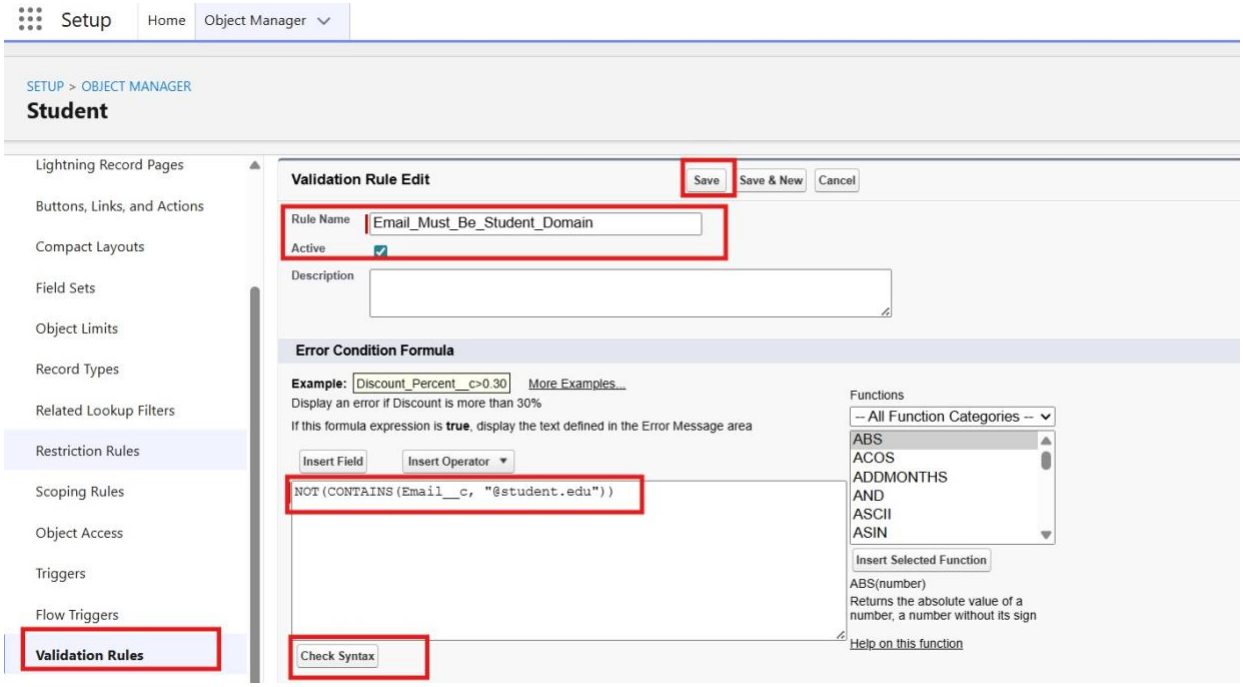
Purpose:
To ensure students use only allowed institutional email addresses.

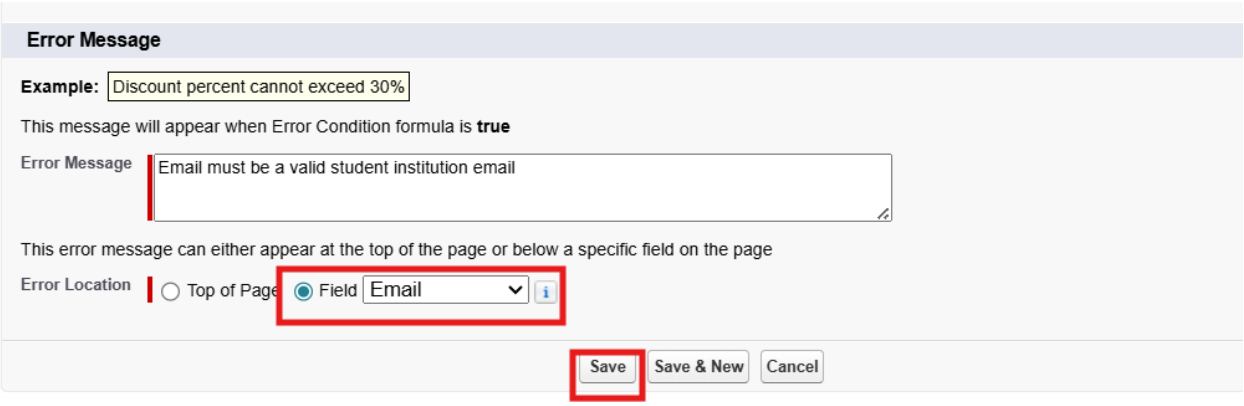
Steps:

Go to Setup → Object Manager → Student__c.
Click Validation Rules → New.
Rule Name: Student_Email_Domain_Validation.
Error Condition Formula: (example for @student.edu)

- NOT(CONTAINS(Email__c, "@student.edu"))

Error Message: Email must be a valid student institutional email.
Error Location: Field → Email__c.
Click Save.





Activity 2: Fees Must Be Paid Before Approval

Purpose:
Prevent Enrollment from being approved until fees are paid.

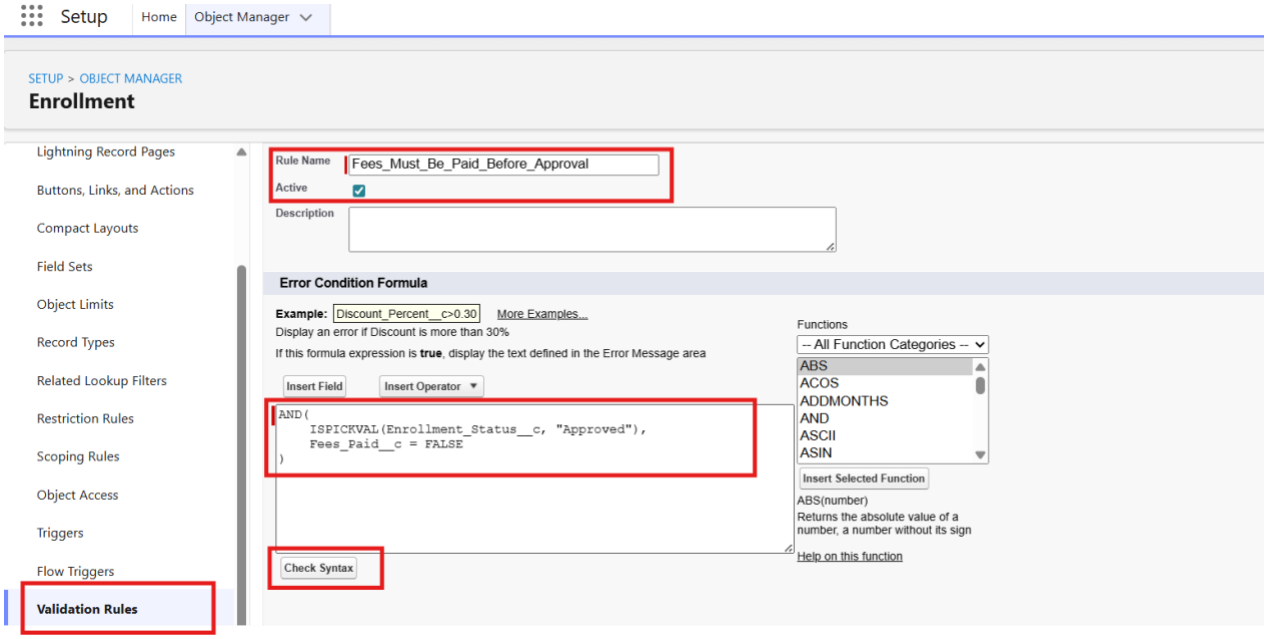
Steps:

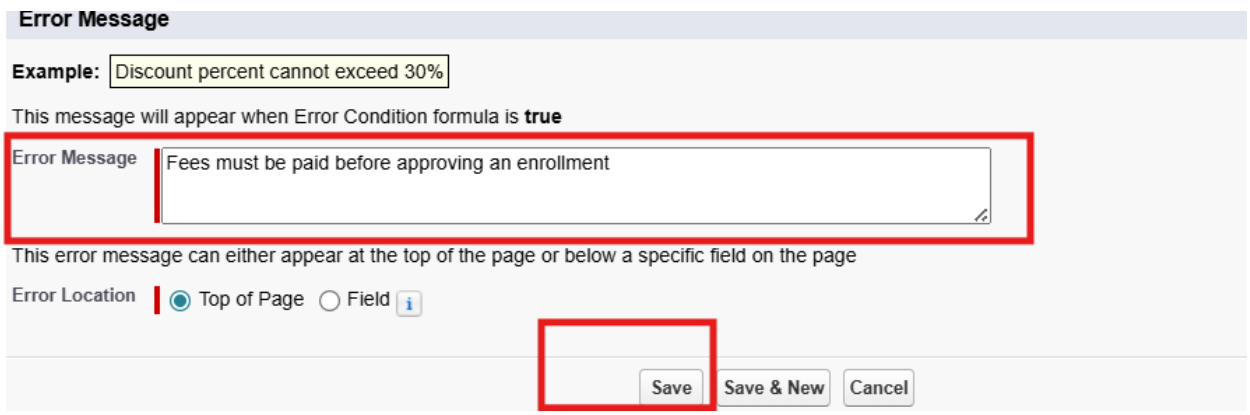
1. Go to Setup → Object Manager → Enrollment__c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:

```
AND(
  ISPICKVAL(Enrollment_Status__c, "Approved"),
  Fees_Paid__c = FALSE
)
```

5. Error Message: Fees must be paid before approving an enrollment.
6. Error Location: Top of Page (or field as needed).

Click Save.





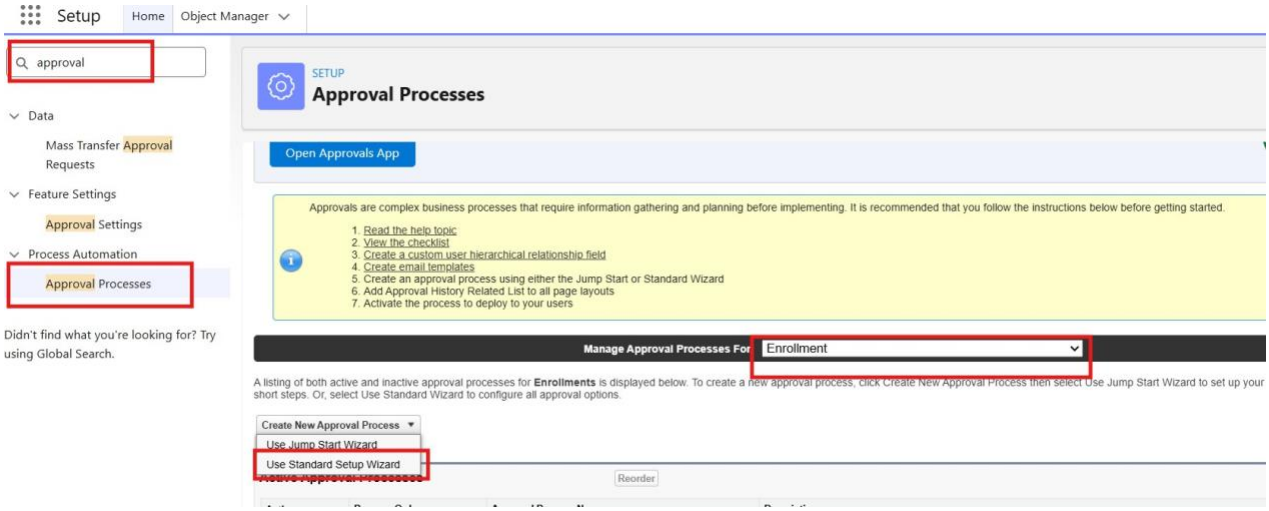
Milestone 6 – Approval Process

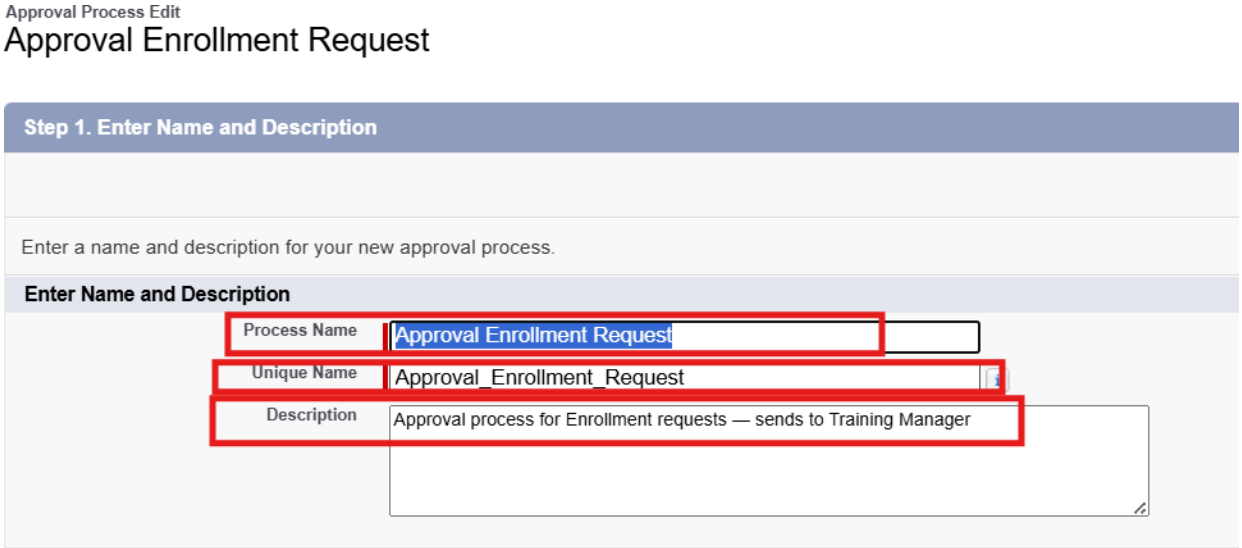
Activity 1: Create Approval Process for Enrollment Requests

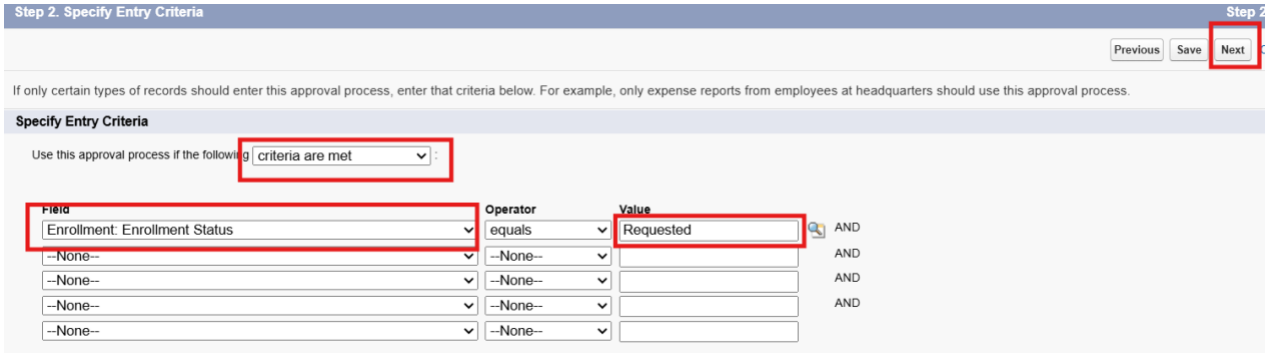
Purpose:
Route Enrollment requests to Training Manager for approval or rejection.

Steps:

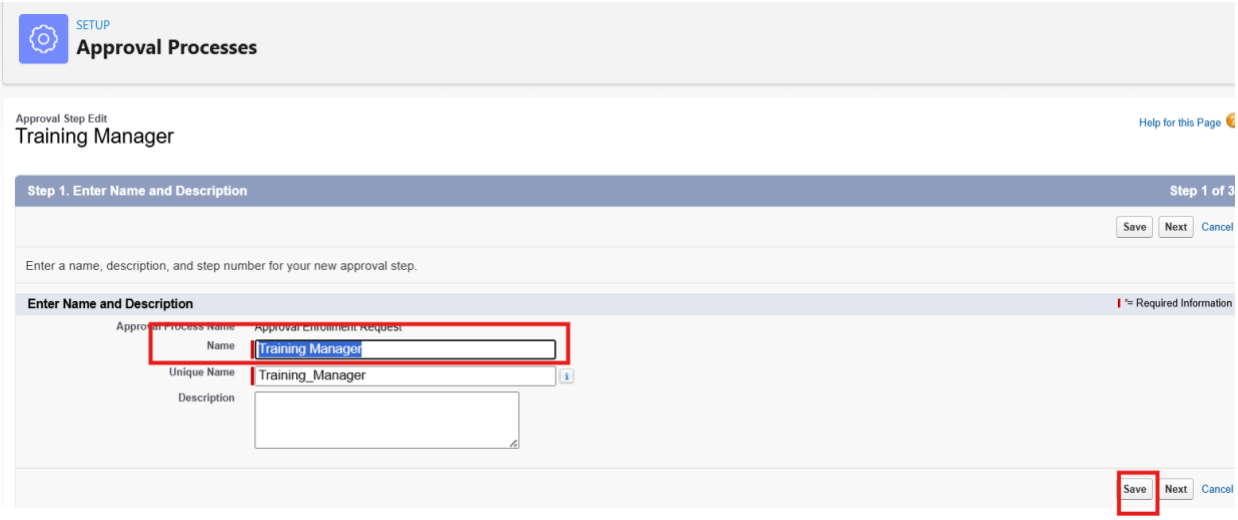
1. Go to Setup → Approval Processes → select Enrollment__c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment_Status__c = "Requested".
5. Save and configure steps.



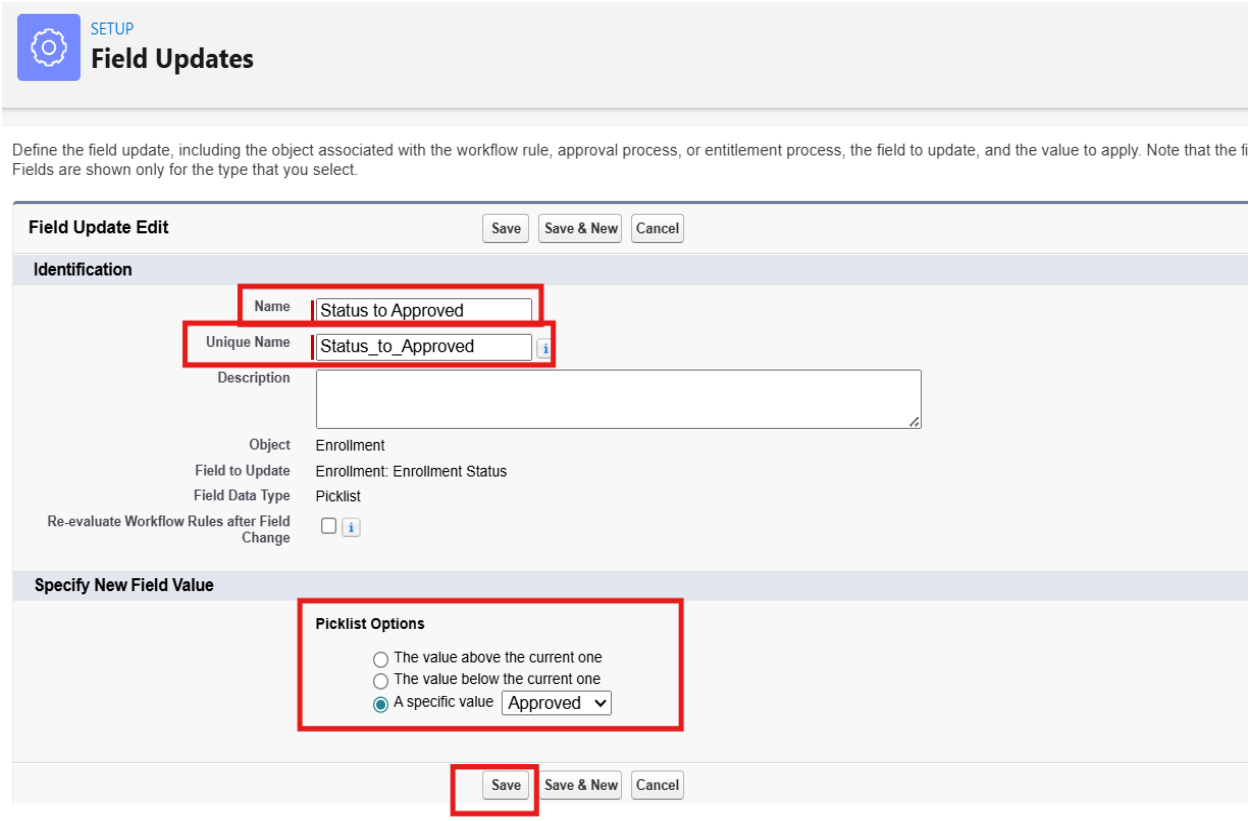




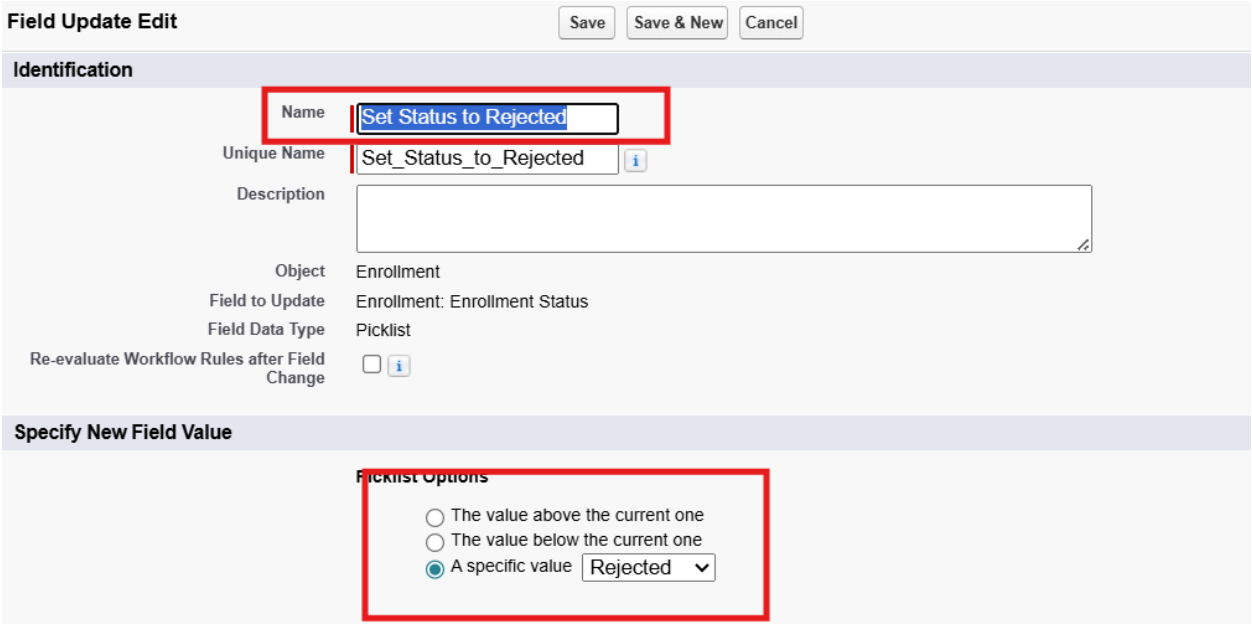
6. Add Approval Step:
7. Approver: specific Training Manager user.



8. Under Final Approval Actions:
9. Add Field Update to set Enrollment_Status__c = "Approved".



10. Under Final Rejection Actions:
11. Add Field Update to set Enrollment_Status__c = "Rejected".



12. Add Email Alert to send rejection email to student.

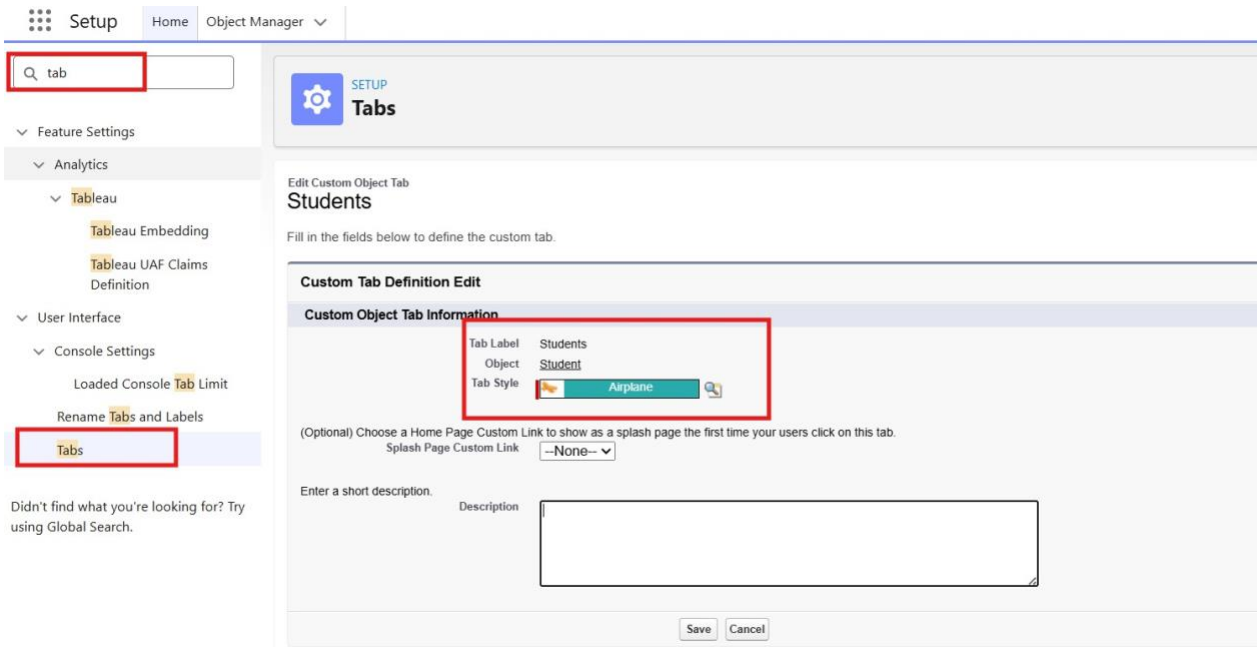
13. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.
4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.



Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.
Follow the same steps as mentioned in Activity -1

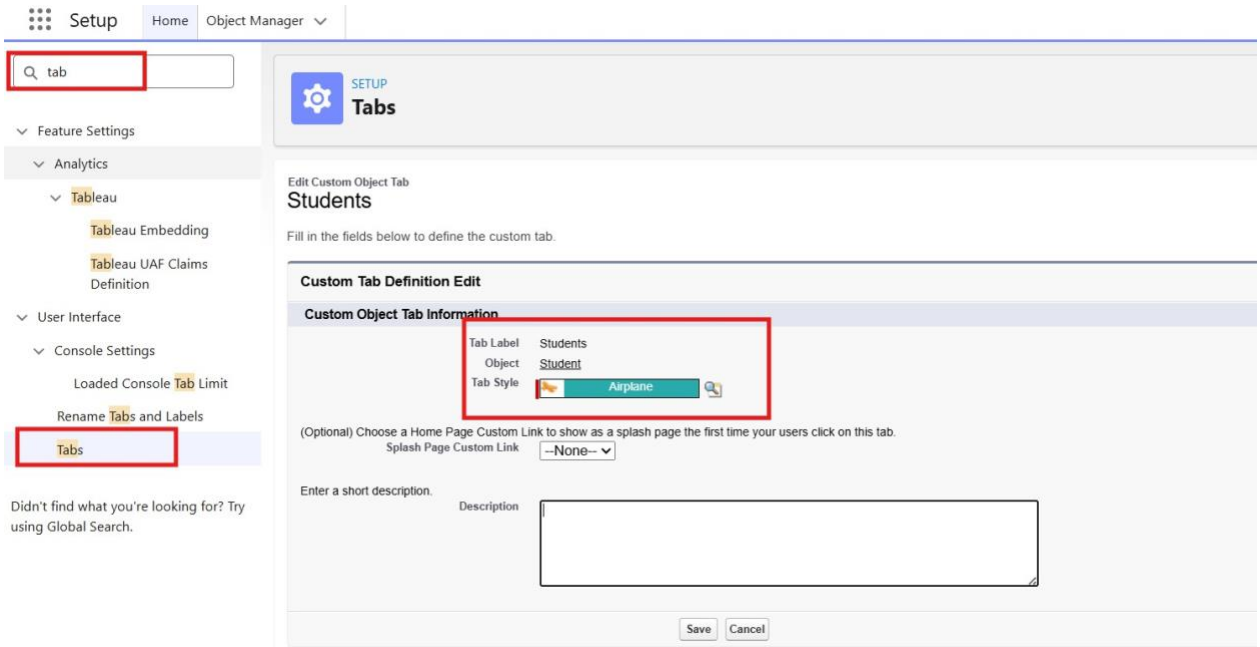
Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box.
Click New Lightning App.

1. Choose a tab style (icon).
2. Click Next, assign it to all profiles, and then click Save.



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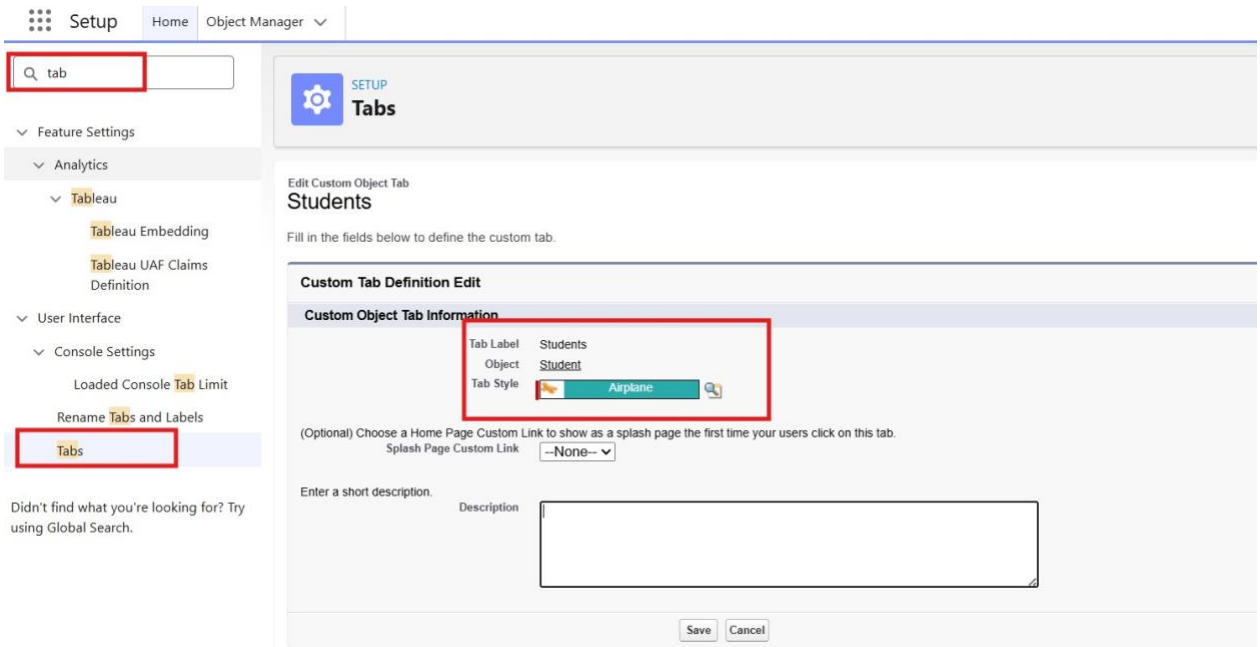

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Milestone 8 - The Lightning App

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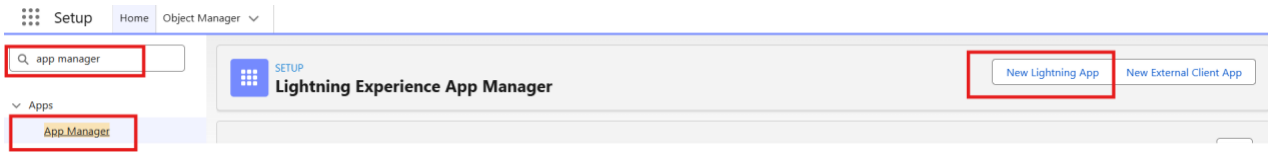
Navigate to Setup.

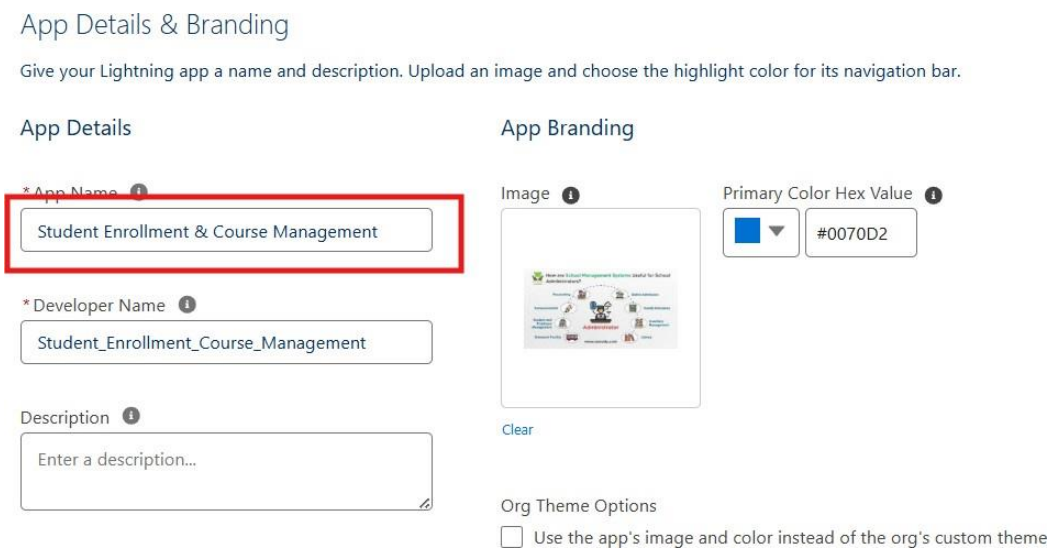
Enter App Manager in the Quick Find search box.
Click New Lightning App.

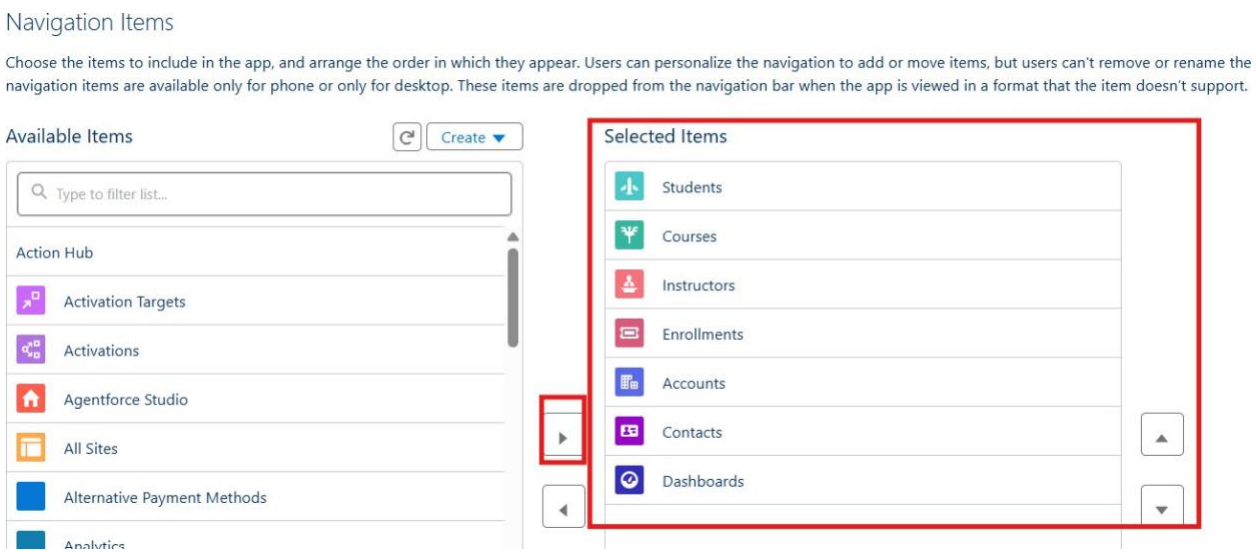
Provide the app details:
App Name: Student Enrollment & Course Management
Developer Name auto-fills
Description: Optional, but can describe the purpose of the app
Click Next and choose the following settings:
App Branding (optional): Add logo or select color theme
Navigation Style: Choose Standard Navigation
Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).
Click Next to select the items (tabs) to be included in the app. Add:

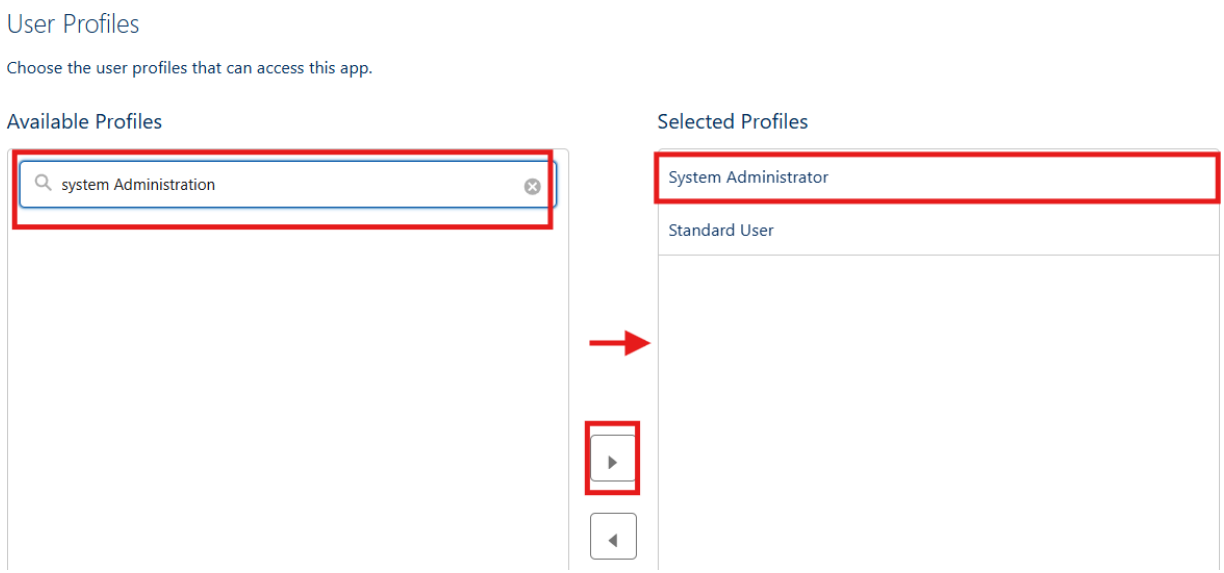
1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.
Finally, click Save & Finish.







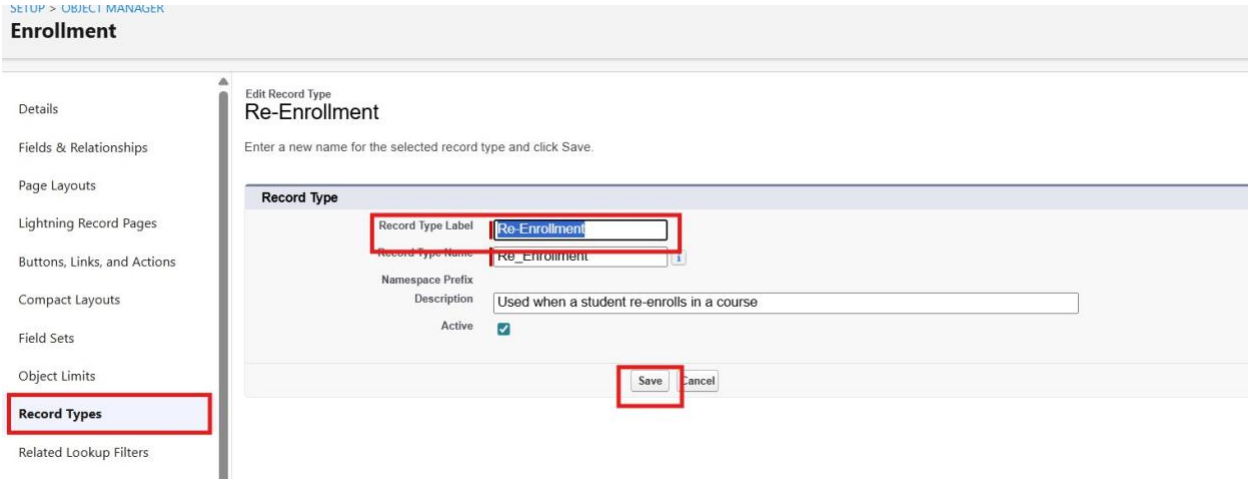


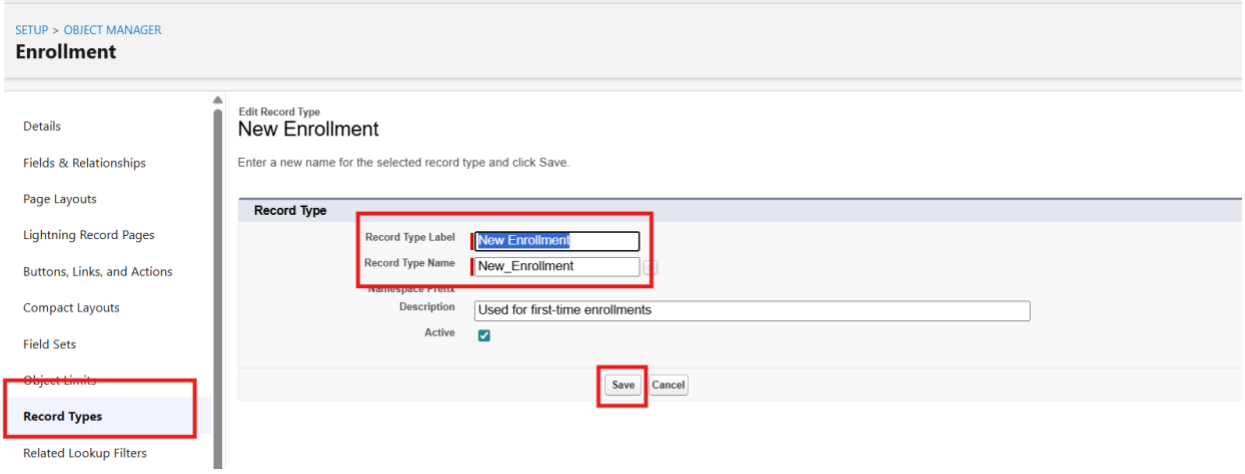
Milestone 9 – Page Layouts and Record:

Activity 1: Create Record Types for Enrollment:

Purpose:
To separate New Enrollment vs Re-Enrollment processes.
Steps:
1. Go to Setup → Object Manager → Enrollment__c.

2. Click Record Types → New.
3. For the first record type:
   - Existing Record Type: Master
   - Record Type Label: New Enrollment
   - Enable for required profiles (Admin, Enrollment Officer).
   - Select the default Page Layout → Save & New.
4. For the second record type:
   - Record Type Label: Re-Enrollment
   - Enable for same profiles.
   - Select the default Page Layout → Save.



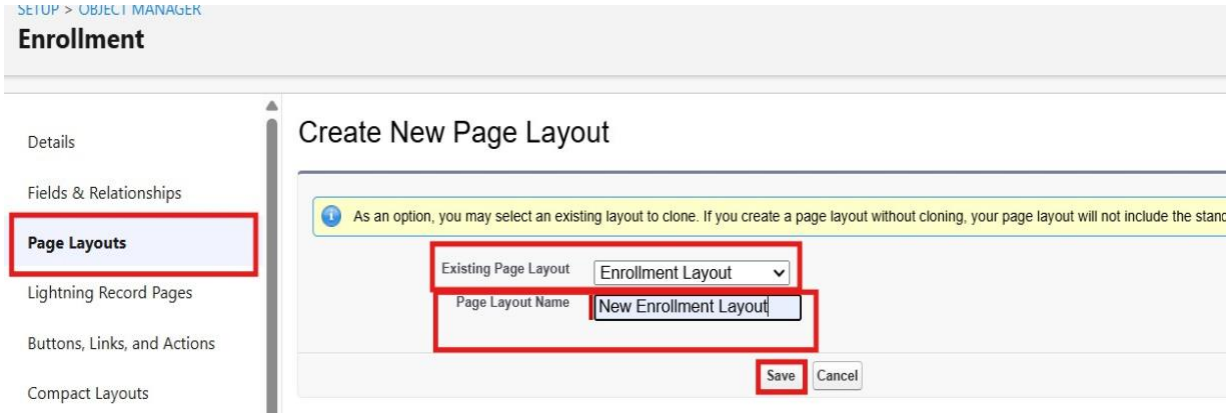


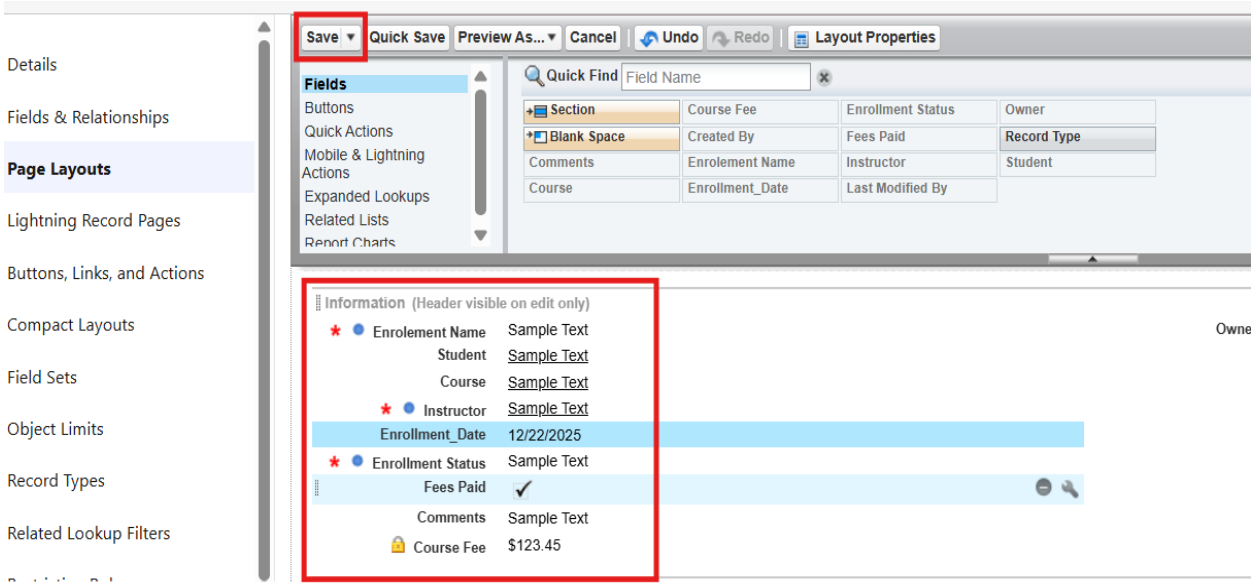
Activity 2: Separate Page Layo Createuts for Each Enrollment Record Type

Purpose:
To show different fields for New Enrollment vs Re-Enrollment.

Steps:

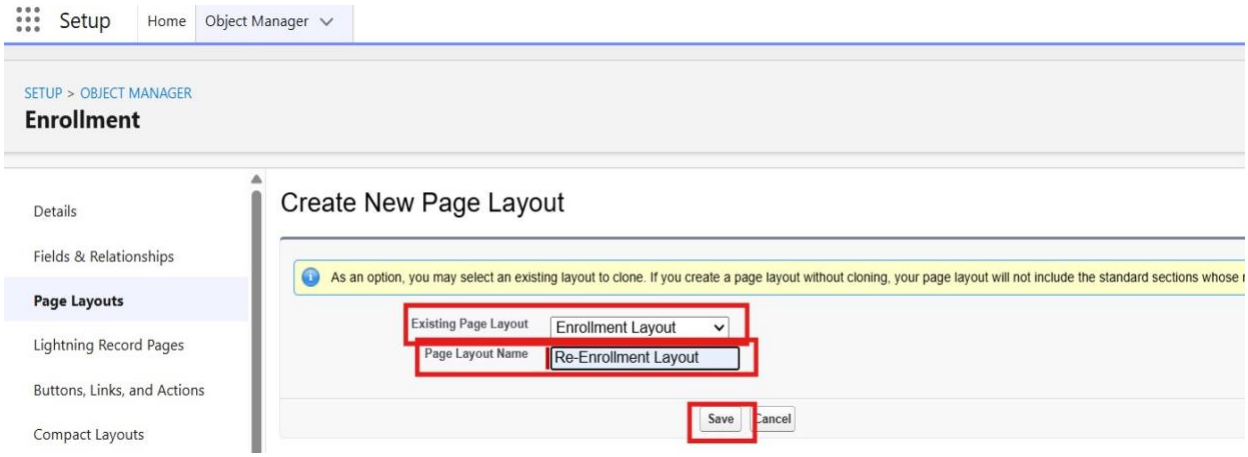
1. Enrollment__c object → click Page Layouts.
2. Click New to create a layout for New Enrollment:
   - Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
   - Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
   - Remove unnecessary re-enrollment fields if any.
   - Click Save.

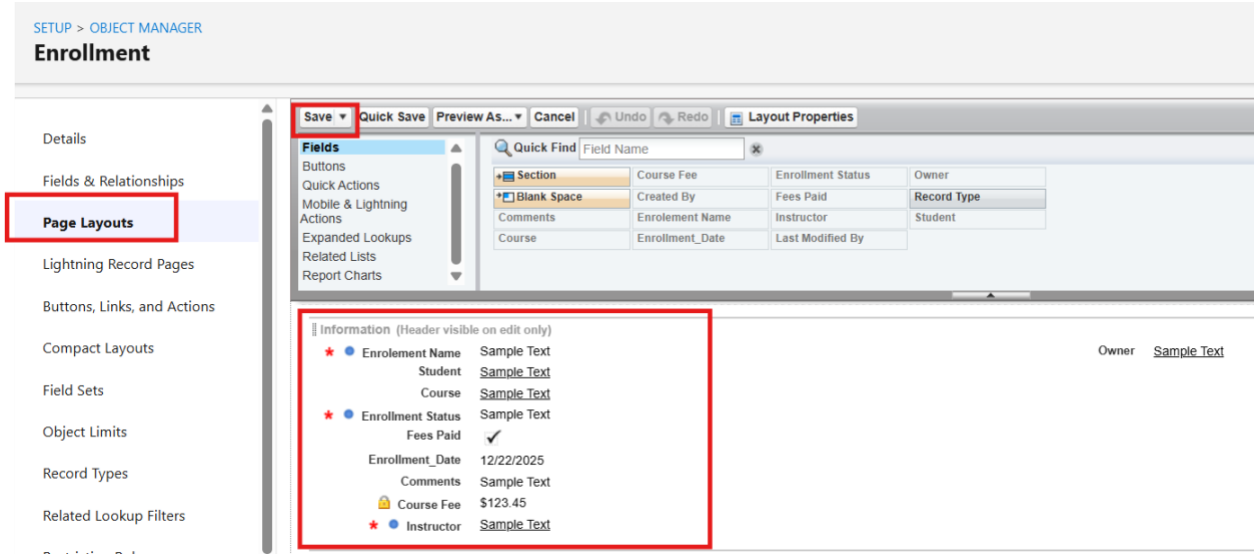




3. Again, click New to create Re-Enrollment Layout:

- Clone existing layout → name it Re-Enrollment Layout → Save.
- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.





4. Click Page Layout Assignment → Edit Assignment.
5. For each profile and record type:

- New Enrollment → use New Enrollment Layout.

6. Re-Enrollment → use Re-Enrollment Layout.

Click Save.

Activity 3: Enrollment R Addelated Lists to Parent Objects:

Purpose:
To see all Enrollments from Student, Course, and Instructor records.
a) Student → Enrollment Related List
Go to Setup → Object Manager → Student__c.
Click Page Layouts → open the main layout.
In the palette, select Related Lists.
Drag Enrollments related list onto the layout.
Customize columns if needed → Save.
b) Course → Enrollment Related List
Go to Object Manager → Course__c → Page Layouts.
Open layout → add Enrollments related list.
Save.
c) Instructor → Enrollment Related List
Go to Object Manager → Instructor__c → Page Layouts.
Open layout → add Enrollments related list.
Save.

Milestone 10 – Automation with Flow

Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

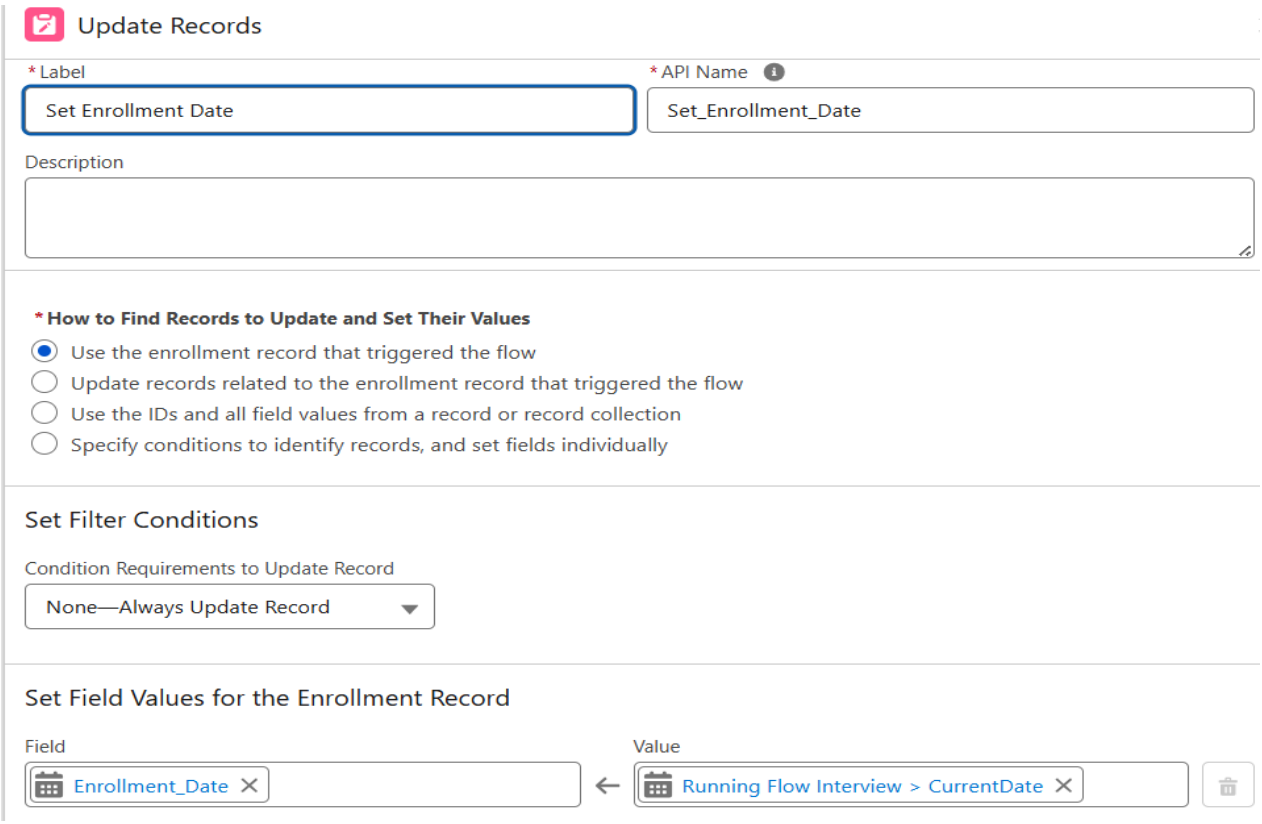
Purpose:
Set Enrollment_Date__c to today when Enrollment is created.

Steps:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment__c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.

6. Add Update Records element:

- Label: Set Enrollment Date.
- Use record that triggered the flow.



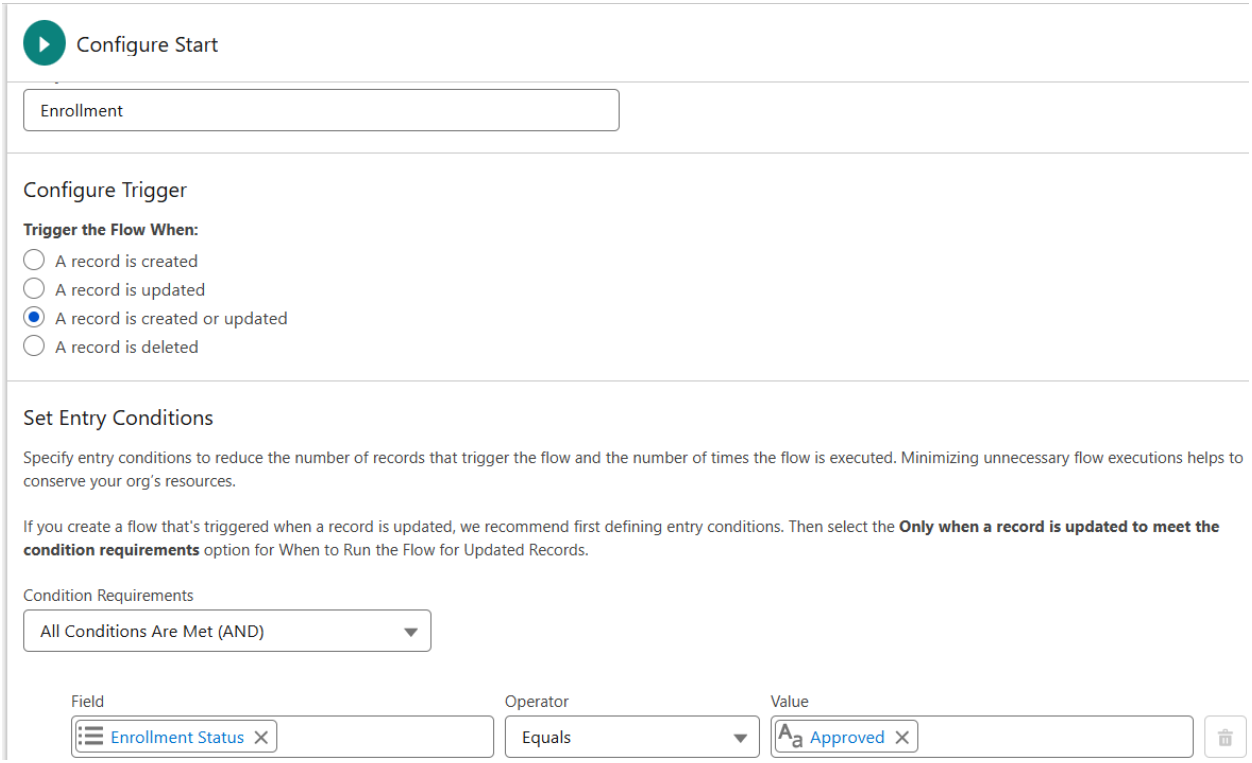
- Field: Enrollment_Date__c → Value: $Flow.CurrentDate.

7. Save as SetEnrollmentDate → Activate.

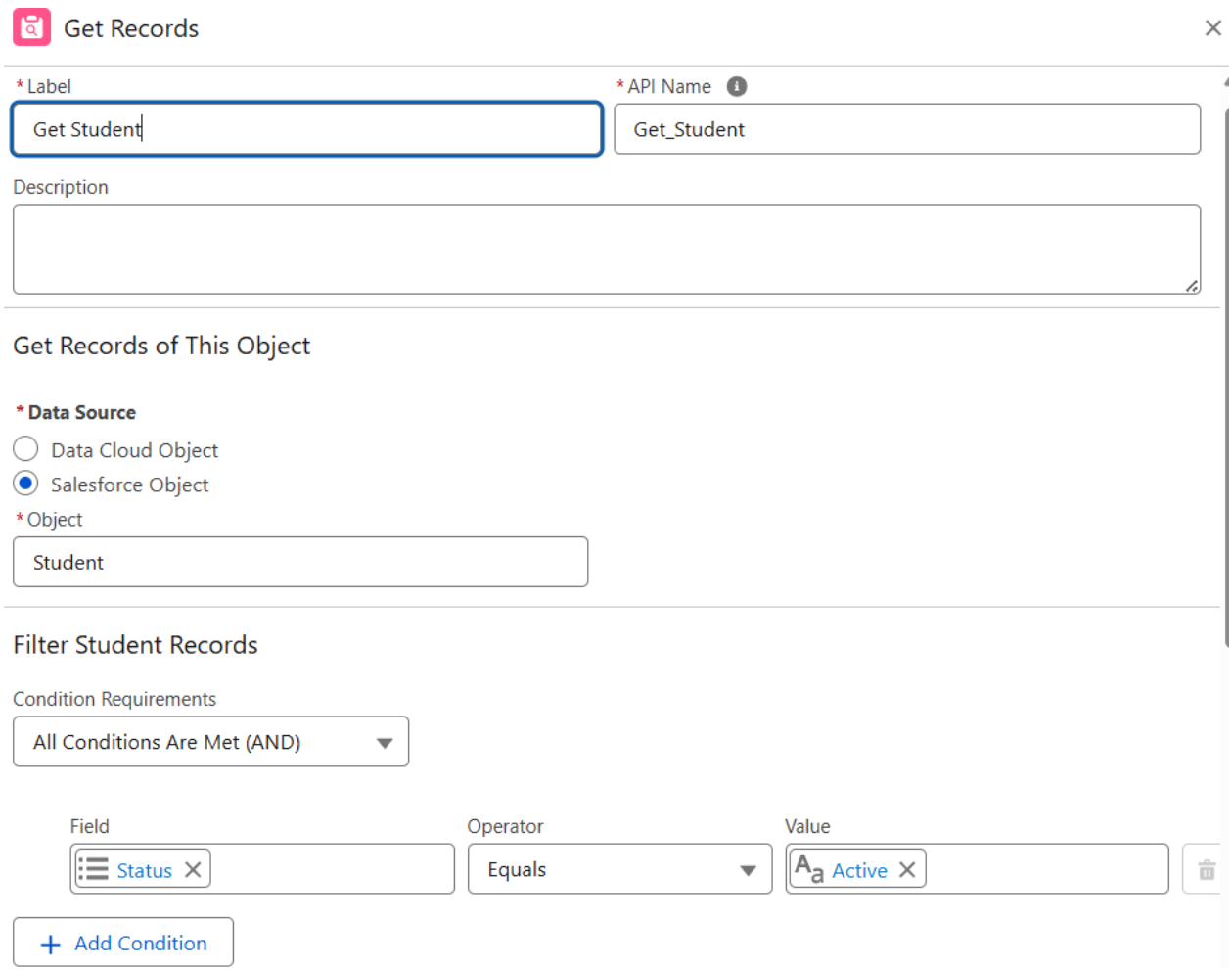
Activity 2: Send Email When Enrollment Approved

Purpose:
When Enrollment_Status__c = Approved, send email, alert admin, and set Student to Active.
Steps (high level for document):

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment__c, Trigger on update.
4. Condition: Enrollment_Status__c Equals "Approved".



5. Add Get Records to fetch related Student__c.



6. Add Action (Email Alert) to send email to student.

Enrollment Approved EmailAlert

* Label

Send Student Email

* API Name

Send_Student_Email

Description

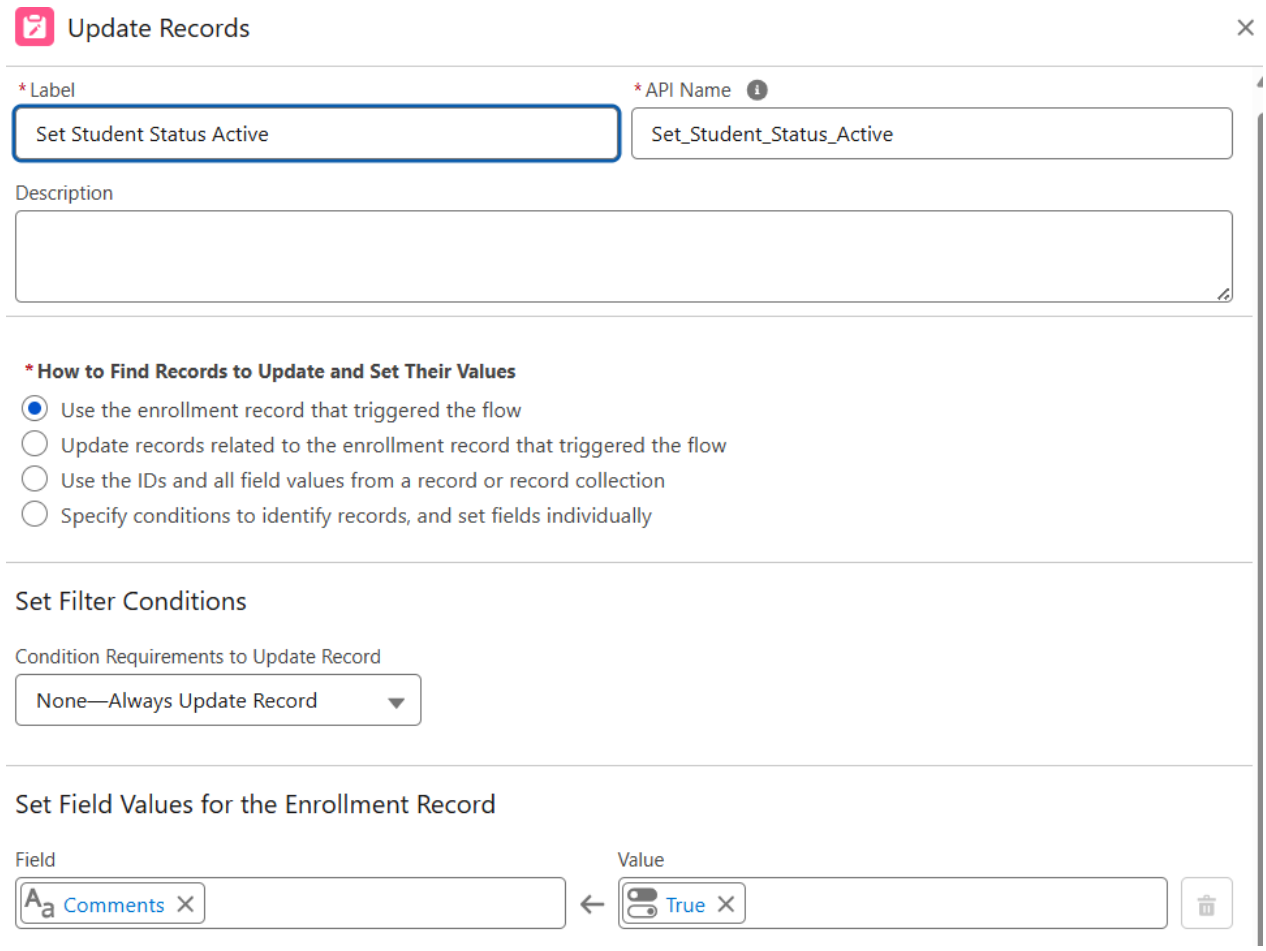
Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

* Record ID

Triggering Enrolement__c > Record ID

7. Add Action to notify Admin if needed.
8. Add Update Records to set Student__c.Status__c = "Active".



9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

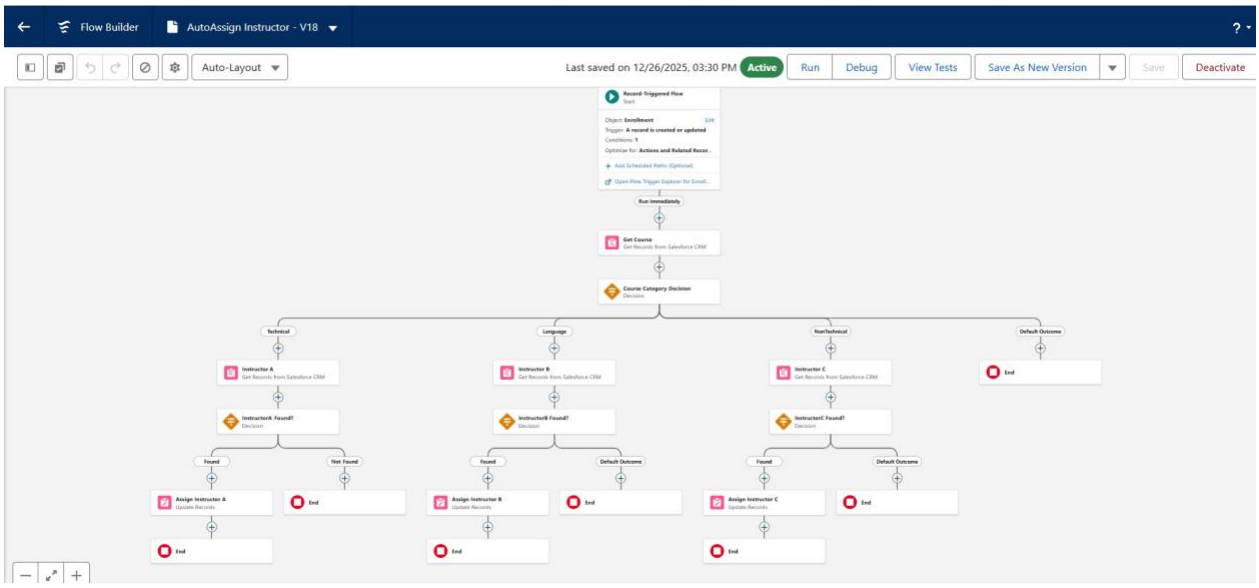
Purpose:
Assign Instructor automatically based on Course category.

Rules:

Technical → Instructor A
Language → Instructor B
Non-Technical → Instructor C

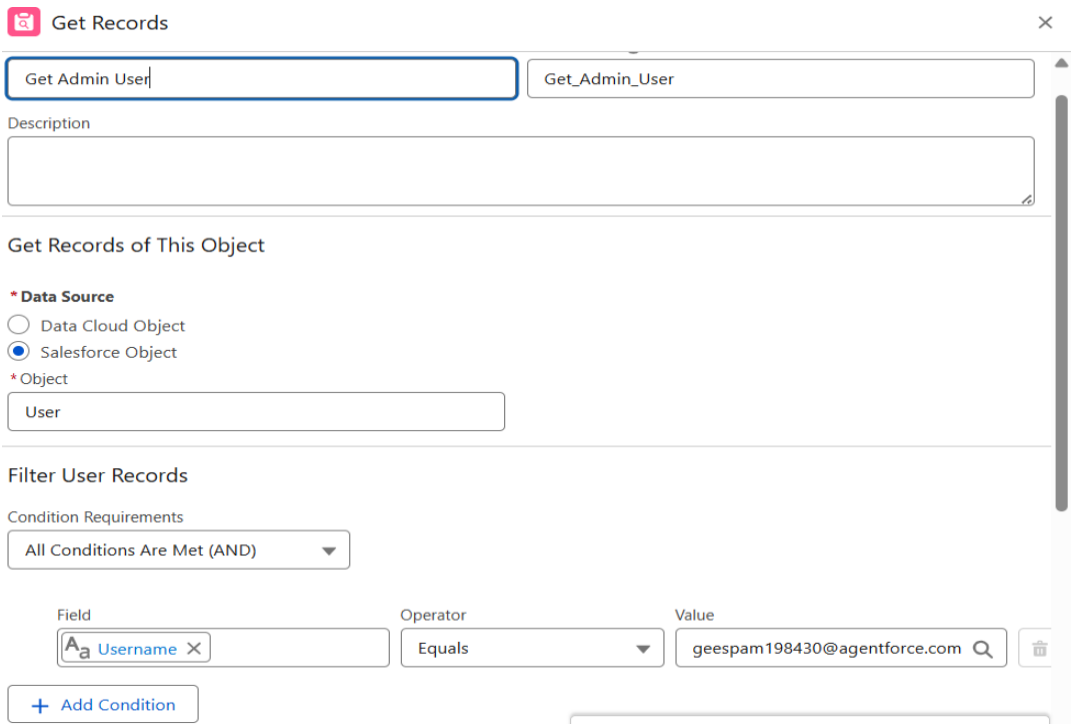
Steps:

1. Prepare Instructor__c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment__c (created/updated).
3. Add Get Records to fetch related Course__c.
4. Add Decision element on Course__r.Category__c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor__c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.

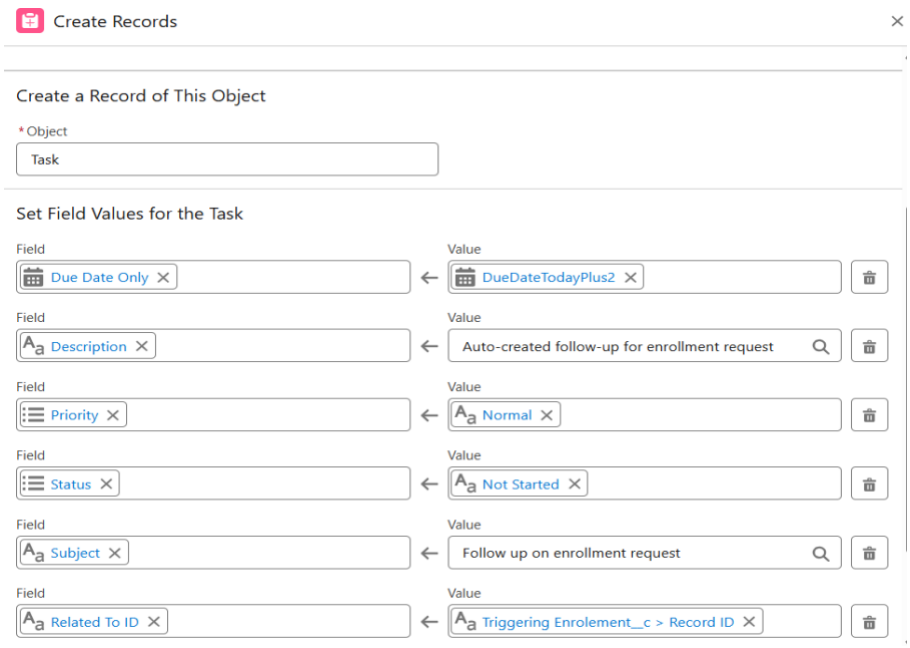


Activity 4: Auto-Create Task for Follow-Up

Purpose:
When Enrollment is “Requested”, create a follow-up task for Admin.



1. Add Create Records for Task:
2. Subject: Follow up on enrollment request.
3. OwnerId: Admin user Id.
4. WhatId: Enrollment Id.
5. ActivityDate: formula Today + 2.



6. Save as Create_Followup_Task → Activate.

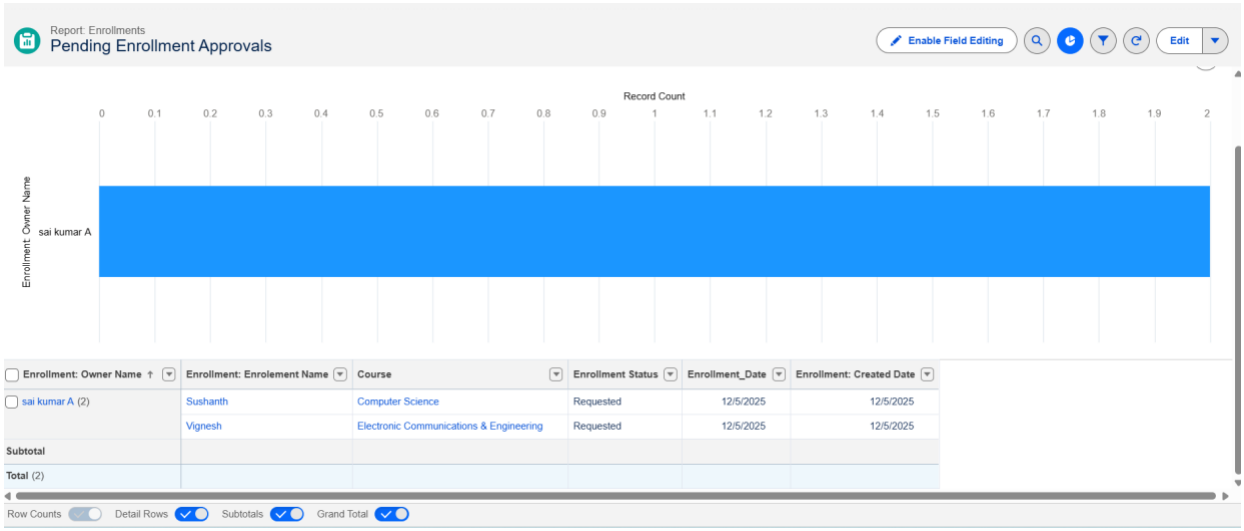
Milestone 11 – Reports:

Activity 1: Create “Students by Status” Report

1. Go to **Reports** → **New Report**.
2. Report Type: `Students`.
3. Group rows by `Status__c`.
4. Add columns as needed (Name, Email, Phone).
5. Save as `Students by Status`.

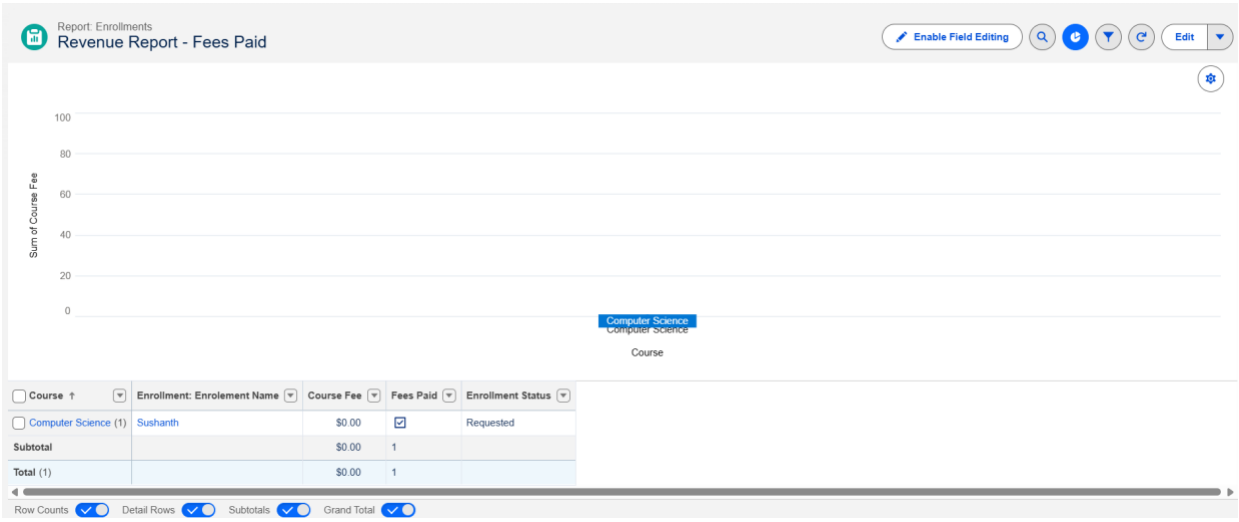
Activity 3: Create “Pending Enrollment Approvals” Report

```
1.New Report → Enrollments report type.
2.Filter: Enrollment_Status__c = "Requested".
3.Add columns (Student, Course, Owner, Created Date).
4.Save as Pending Enrollment Approvals.
```



Activity 4: Create "Revenue Report (Fees Paid)" Report

1. New Report → Enrollments report type.
2. Filter: Fees_Paid__c = TRUE.
3. Group by Course__r.Name.
4. Summarize Course_Fee__c or similar fee field.
5. Save as Revenue Report – Fees Paid.

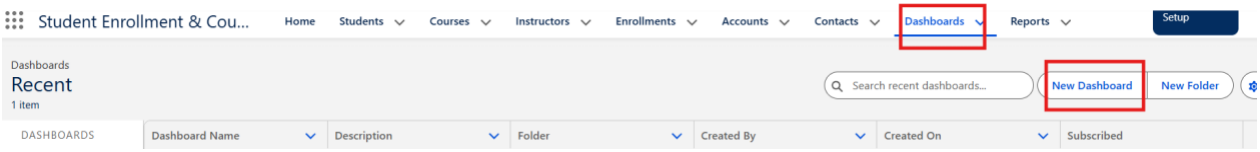


Milestone 12 – Dashboards:

Activity 1: Create “Student Managemet Dnashboard”

Steps:

1. Go to **Dashboards** → **New Dashboard**.
2. Name: Student Management Dashboard.



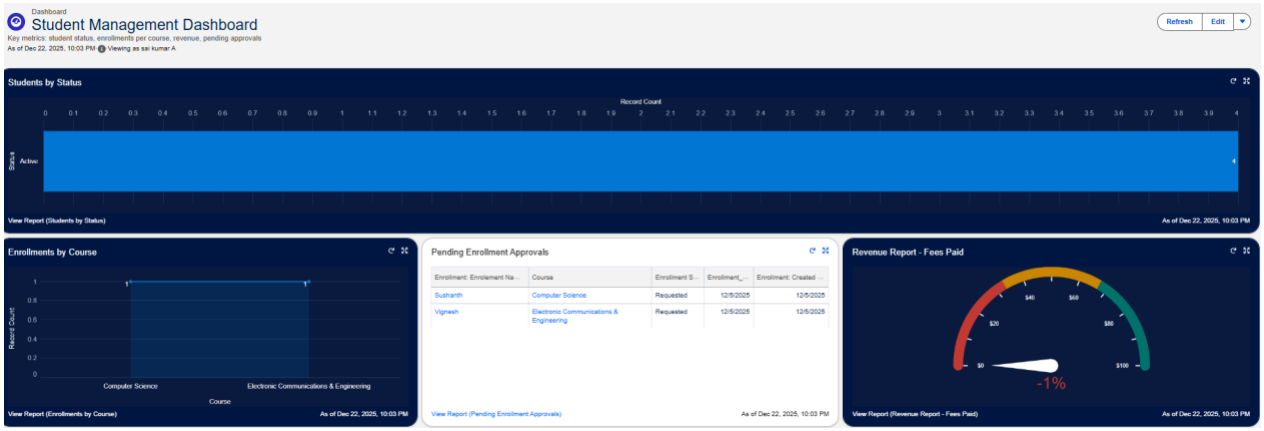
3. Select folder → **Create**.
4. Select new Dashboard folder and Save.

Add components:

- **Donut Chart** → using Students by Status report.
- **Bar Chart** → using Enrollments by Course report.

- **Gauge** → using Revenue Report – Fees Paid (sum of revenue field).
- **Table** → using Pending Enrollment Approvals report.

4. Arrange components on canvas.



THANKYOU